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Player-1: Learning to Play from Synthetic Experience

Extending the VLM Action Space with Relative Motion, Sustained Inputs, and Composite Actions, then Training with PPO in 16 Synthetic Environments, Triples MineDojo-300 Success and Transfers to Held-Out Games

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Standard computer-use agents fail at digital gameplay not because their policies are too weak, but because their action spaces cannot express the motor primitives games require: relative pointer motion, sustained directional inputs, and composite simultaneous key combinations. *Player-1* fixes the representation gap first: it extends the GUI action vocabulary, fine-tunes on 6,044 curated gameplay trajectories (SFT), then runs PPO in Player-1-Gym—16 synthetic 2D and 3D environments spanning five control families. Player-1-9B raises MineDojo-300 success from 12.63% (best prior baseline) to **38.33%**; gains transfer to ViZDoom-10 and GameWorld without any game-specific fine-tuning.

AT A GLANCE

Problem: Standard GUI action spaces lack relative pointer motion, sustained inputs, and composite key combos—the primitives digital games require; agents fail not from limited policy but from limited vocabulary.

Why it matters: Game-playing is a rich testbed for continuous motor-skill learning in VLMs; solving the action-space gap is a prerequisite for any game-capable generalist agent.

Method: Extend the action space; SFT on 6,044 curated trajectories; PPO in Player-1-Gym (16 procedurally generated 2D/3D environments, 5 control families: aiming, combat, driving, navigation, UI).

Result: Player-1-9B: 38.33% on MineDojo-300 (+25.7 pp over best prior); 35B-A3B reaches 41.04%; transfer to held-out ViZDoom-10 and GameWorld.

Evidence: MineDojo-300, ViZDoom-10, GameWorld; 9B and 35B-A3B; SFT-only ablation.

The Big Picture

Vision-language models now navigate browsers, fill forms, and operate desktop applications with reasonable accuracy. Interactive games present a qualitatively different challenge: staying alive in a first-person shooter, steering a vehicle, executing a combat combo, or sprinting while turning requires motor skills— sustained, feedback-driven, multi-axis control—that desktop GUI tasks never demand.

The failure mode is not a policy problem. If the action space cannot express "move the mouse 12 pixels right while holding the left mouse button," no amount of policy optimization will produce that action. Player-1 diagnoses this as a representation gap and closes it before applying RL.

Why Existing Approaches Fall Short

SFT-only agents can imitate successful play trajectories, but imitation is brittle to novel game states and does not generalize to games outside the demonstration distribution.

Standard GUI agents treat games like web navigation: they produce discrete, stateless click-and-keystroke actions. They cannot execute the sustained directional inputs needed for driving, the relative pointer sweeps needed for aiming, or the simultaneous multi-key presses needed for combat.

Game-specific RL agents achieve strong results on individual titles (Atari, StarCraft) but require game-specific reward shaping, environment wrappers, and action-space engineering; they do not transfer to new games or VLM-based GUI architectures.

Real-game RL training requires running live game executables: it is slow, platform-dependent, and legally constrained for commercial titles.

Core Mechanism: Extended Action Space and Player-1-Gym

Extended action space:

$$\mathcal{A}_{\text{ext}} = \mathcal{A}_{\text{std}} \cup \mathcal{A}_{\text{rel}} \cup \mathcal{A}_{\text{hold}} \cup \mathcal{A}_{\text{combo}}$$

where $\mathcal{A}_{\text{rel}}$ adds relative pointer motions $(\Delta x, \Delta y) \in \mathbb{R}^2$, $\mathcal{A}_{\text{hold}}$ adds hold/release primitives for keys and buttons, and $\mathcal{A}_{\text{combo}}$ adds composite simultaneous key-plus-pointer sequences.

Player-1-Gym is a suite of 16 synthetic 2D and 3D environments covering five control families:

1. Aiming (first-person pointer control)
2. Combat (sustained + combo actions under adversarial pressure)
3. Driving (continuous relative steering)
4. Navigation (spatial orientation + locomotion)

5. UI interaction (menus, inventory, crafting)

Environments are procedurally generated so each episode presents a novel layout, preventing simple memorization of visual patterns.

Technical Formulation

The SFT phase minimizes the standard next-token prediction loss over a demonstration dataset $\mathcal{D} = \{(o_t, a_t)\}_{t=1}^{T}$ annotated with extended actions:

$$\mathcal{L}_{\text{SFT}}(\theta) = -\mathbb{E}_{(o_t,a_t)\sim\mathcal{D}}[\log \pi_\theta(a_t \mid o_t)]$$

The PPO phase optimizes the clipped surrogate objective with advantage estimates computed via GAE:

$$\mathcal{L}_{\text{PPO}}(\theta) = \mathbb{E}_t\Big[\min\Big(r_t(\theta)\hat{A}_t,\ \text{clip}(r_t(\theta), 1{-}\varepsilon, 1{+}\varepsilon)\,\hat{A}_t\Big)\Big]$$

where the probability ratio is $r_t(\theta) = \pi_\theta(a_t \mid o_t)/\pi_{\theta_{\text{old}}}(a_t \mid o_t)$ and $\hat{A}_t$ is the generalized advantage estimate.

The value function $V_\phi(o_t)$ is trained simultaneously with

$$\mathcal{L}_V(\phi) = \mathbb{E}_t\left[\left(V_\phi(o_t) - \hat{R}_t\right)^2\right]$$

where $\hat{R}_t = \sum_{k=0}^{T-t} \gamma^k r_{t+k}$ is the discounted return.

Reward signals for Player-1-Gym are binary task-success indicators plus progress-based shaped rewards derived from task predicates.

Learning or Inference Procedure

1. **SFT:** Fine-tune the VLM on 6,044 human gameplay trajectories annotated with $\mathcal{A}_{\text{ext}}$ labels.
2. **Deploy to Player-1-Gym:** Instantiate 16 environments with procedurally generated episodes; set episode timeout.
3. **PPO rollout:** Asynchronous workers generate episodes; a central policy-update server applies gradient steps every N rollout steps.
4. **Evaluation:** Zero-shot inference on MineDojo-300, ViZDoom-10, and GameWorld using the RL-trained policy; no game-specific fine-tuning.

What the Guarantee Says

Player-1 establishes the following empirical claims across model sizes and held-out benchmarks:

- Player-1-9B: MineDojo-300 success = 38.33% vs. 12.63% for the best prior baseline (+25.7 pp).
- Player-1-35B-A3B: MineDojo-300 success = 41.04% (+17.7 pp over its own SFT policy).
- Zero-shot transfer to ViZDoom-10 and GameWorld (both outside Player-1-Gym); both show positive gains over the SFT-only policy.
- SFT-only policy scores substantially below the PPO-trained model on all evaluated benchmarks, confirming RL drives the improvement.

Experimental Findings

- **MineDojo-300:** Player-1-9B: 38.33%; Player-1-35B-A3B: 41.04%; best prior: 12.63%.
- **ViZDoom-10:** Positive transfer; largest gains on aiming and combat subtasks that most require relative-motion actions.
- **GameWorld:** Positive transfer across navigation and UI subtasks.
- **SFT ablation:** SFT-only policy (no PPO) shows substantial gap below the full Player-1 model on all three benchmarks.
- **Control family breakdown:** Aiming and combat show the largest absolute gains, confirming that the new action primitives are the key bottleneck for those task families.

Ablations and Interpretation

Action space ablation: Removing $\mathcal{A}_{\text{rel}}$ from the action set causes a large drop on aiming and driving tasks; removing $\mathcal{A}_{\text{combo}}$ degrades combat performance.

Gym environment count: Performance on MineDojo-300 improves as the number of Player-1-Gym environments increases; the diversity of motor tasks covered—not just quantity—determines transfer breadth.

SFT warmup importance: Removing the SFT phase and training RL from the base VLM significantly slows convergence; the SFT demonstration data provides a useful initial policy distribution that shortens RL exploration.

Model scale: 35B-A3B outperforms 9B (+2.71 pp on MineDojo-300); the gap is larger on ViZDoom, suggesting complex first-person reasoning benefits more from scale than pure motor-control tasks.

Why transfer works: The five control families in Player-1-Gym cover the motor primitives common to most PC games; RL-acquired motor policies generalize across visual appearances and reward structures that differ from training, because the underlying motor sub-problem is shared.

Reference

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